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TASK RISK ASSESSMENT (TRA) GUIDELINE

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1.0 PURPOSE

This Document is intended to guide the Idd El Shargi personnel in how to execute the Task Risk Assessment (TRA) technique in line with the Idd El Shargi Procedure for Process Hazard review, required in IS-HSE-PRC-308. The General objectives of the Idd El Shargi Risk Management Guidelines (RMG), within the Idd El Shargi RM process are to:

- Allow a clearer definition of requirements with regard to the Idd El Shargi RM program and related Policies and Procedures.
- Allow specific topics to be addressed in detail, relevant to the application of the RM element, including examples where relevant; and
- Act as a basis for further information, understanding and training.

Note: These Documents should be considered as Guidelines only and not mandatory unless otherwise stated, the use of the words, 'shall' and 'should' will indicate a mandatory requirement. For further clarification on content and intent contact Idd El Shargi HSE Risk Engineer.

2.0 SCOPE AND APPLICABILITY

The Guideline applies to all personnel within Idd El Shargi plants and facilities (PS-1, ISND, ISSD and ISHT), who are involved in Risk Management, and applies to Employees, contractors, and subcontractors, whether engaged directly by Qatar Energy or by others, who will perform work or services of any kind at any Idd El Shargi facilities and/or locations.

This Guideline covers non-routine transient tasks in all process systems containing hydrocarbons or other hazardous fluids, utilities, safety systems, campaign activities, and MOC related activities which may pose a risk to personnel, the environment, or may lead to property damage and loss of asset and revenue. This Guideline only addresses those aspects which are unique to Task Risk Assessment technique. It does not cover the follow-up activities.

3.0 GUIDELINE

3.1 GENERAL

The TRA is a new term being introduced at Idd El Shargi. The prior term used for these studies was, On-Shore Job Safety Analysis (JSA). The term JSA, however, caused confusion with pre-job safety checks conducted offshore (e.g. tool box talks). A TRA is specifically an onshore based (i.e. Doha office) qualitative risk review using a team-based brainstorming approach.

The TRA technique is used in Idd El Shargi to assess risk in procedural type activities, i.e. job activity based risk. A TRA is a systematic analysis of a specific job that:

- Identifies any hazards associated with the job tasks.
- Determines controls to mitigate the risks associated with the hazards identified.
- Confirms the correct method and conditions to perform a job safely.

Typical task covered by TRA are as follows:

- Tasks posing risk of unplanned equipment / system / plant trip.
- Suspended / cantilevered work from a fixed structure.
- Isolation of high voltage equipment (1,000 V or more).
- Work in difficult access areas (at height, over the side, mobile elevated platforms).
- Work with radioactive materials.
- Work involving excavation or trenching deeper than 40 cm.
- Work involving use of mechanical excavation equipment.
- A new task not previously carried out at Idd El Shargi.

The team-based technique requires that specific work activities are broken down into task elements, and procedural steps identified for each task. Potential incidents or events are then identified against each step. The subsequent risk is evaluated, and additional controls identified for risk reduction. The change owner is responsible for ensuring that the activity is broken down into suitable tasks and corresponding steps in advance of the assessment. Types of risks under TRA focus are the following:

- Inherent risks: risks associated with design of workplace, its equipment and location.
- Process-related risks: risks arising from the process being carried out, fluid properties and process conditions.
- Task-related risks: caused by the activities of people in the workplace.

As well as a task/ stepwise breakdown, the details of equipment, personnel, materials and fluids involved in the activity should be clearly identified for the review. Key elements of the review include:

- Potential Incidents from the task step.
- Consequences, impact to PAF, PD/ LOR, and ER.
- Intrinsic Controls (see TRA Appendix 3 for a specific list).
- Risk Ranking based on Idd El Shargi TRA Risk Matrix.
- Additional Controls/ Recommendations/ Request to Worksite.
- Residual Risk.

3.2 TRA REVIEW TEAM

3.2.1 TEAM STRUCTURE

In Idd El Shargi, TRA analysis should be led by an independent TRA analysis facilitator, mandated by the Risk Management Key Personnel Guideline (IS-HSE-GDL-302).

A core TRA analysis team shall consist of:

- Independent Facilitator, supported by a scribe.
- Project/Change (MOC) Proponent.
- Engineering Representative.
- HSE and/or Risk Engineering Representative.

- Operations Representative.
- Maintenance Representative.

Team members who may be called upon as required include:

- Asset Integrity Representative
- Construction Representative
- Marine Specialists
- Drilling/ Well Service Engineers
- Specialist Engineering Representatives (Technical Authorities)
- Consultant/Third Party Specialists
- Vendors Representatives
- Others as required

It is essential that the team composition be appropriate for the task being studied. In general, the role of team members is to creatively share their collective operating experiences and process knowledge to:

- Identify credible potential incidents of the task step for discussion.
- Identify and challenge effectiveness of existing safeguards, (e.g. physical safeguards in place, Procedure, expert involvement, etc.).
- Qualitatively evaluate risk of each identified scenario using Idd El Shargi TRA Risk Matrix.
- Make appropriate recommendations aimed at reducing risk.

All core team members attending a TRA analysis should ideally be in attendance for its full duration, without substitution.

3.2.2 TRA FACILITATOR

The TRA facilitator should be experienced personnel with relevant knowledge in occupational safety and/ or production operations in the oil and gas industry. He/she should be experienced, qualified and certified in facilitating TRA studies of a similar level of complexity.

Responsibility for the conduct of the TRA lies with the TRA facilitator. The TRA facilitator has the responsibility to ensure that the TRA study proceeds in an effective and efficient manner, maintaining the essential flow of the study, its robustness and in ensuring the active involvement and participation of all members of the TRA team.

The TRA is recorded on PHA Pro 8 software work sheet, very similar to that used for HAZOP and 'What-if' recording. The TRA Facilitator, with the assistance of the scribe, is responsible for TRA record sheets, for agreement by and acceptance of the team and in preparation of the TRA report.

The TRA Facilitator does not have responsibility for implementation or close out of action items arising from the TRA study.

3.3 METHODOLOGY

3.3.1 REVIEW DEFINITIONS

Preparatory work is required both by the Project/ Change proponent and by the TRA facilitator prior to each session. This work typically comprises:

- Collecting the relevant data, e.g. Safe Work Method Statement, change request Documents, FTIs, work pack, method statement.
- Validation of information.
- Planning the sequence of the analysis, etc.

Once the preparations are completed and the TRA study team assembles, the team should spend a brief period of time to discuss the Purpose, Scope, and Objectives of the analysis. The initial establishment of Purpose, Scope, and Objectives is very important and should be precisely set down so that it will be clear at the time of the study and in the future, of the basic intent. There is a specific sheet to be completed in the Administration section of PHA Pro TRA template for the inclusion of Purpose, Scope, and Objectives.

- The purpose defines the driver behind the analysis, e.g. RM process requirements for a yet to be built project, or facility changes, e.g. using a TRA analysis to review a Procedure for a specific non-routine transient task.

The purpose should take the form of a succinct statement clearly identifying why the TRA is being undertaken.

- The scope of the review is the boundaries of the physical unit, and also the range of events and variables considered. It should also highlight any areas that are not specifically covered.
- The objectives should be distinct statements which expand upon the defined purpose.

In addition, as the Safe Work Method Statement (SWMS), forms the basis for a TRA study, it shall be thoroughly reviewed by the Team prior to ensure thorough understanding and correct structure. SWMS may be in the form of Task Procedure Document or Project Method Statement.

The Project or Change proponent and/ or Task Leader (e.g. Construction representative), should brief the team about important information, including:

- The technical reason or basis for the change.
- Site conditions and potential hazards from the job environment, process equipment, and the task step.
- The expected impact on safety, health, environment, property damage, and potential loss of revenue.
- The intended duration of the task and a time estimate to reinstate the process into normal operations.

3.3.2 TRA REVIEW DOCUMENTS

For MOC type projects and Ad-Hoc reviews, Project or Changes (MOC), the proponent will be responsible for assembling the relevant data to meet the requirements of the Task Risk Analysis Facilitator.

A Safe Work Method Statement (SWMS) shall be issued to all participants for their review and preparation for the TRA prior to the start of the TRA analysis study.

A SWMS specifies all work related tasks and specific steps on how to safely perform each task. This Standard set of steps is considered the most appropriate accepted means of achieving the task. A SWMS should contain sufficient details to be able to analyze the overall task correctly. If the SWMS changes significantly, then the TRA performed is invalid and a revised TRA must be conducted.

The following additional Documents may be required by the TRA analysis team:

- Plot Plans, General Arrangement & Elevation Drawings.
- Site Survey Document (e.g. photographs).
- Material Data Sheets.
- Isolation Pack.
- Procedures and Work Instructions.
- PFDs.
- P&ID's for process and utilities.
- Idd El Shargi TRA Risk Matrix, see Appendix D.
- Typical Control for Tasks, see Appendix E.
- Process Design Basis.

  Etc.

Additional Documents may be called for during the TRA analysis study to amplify or verify information requested by the TRA team.

3.3.3 REVIEW SESSIONS

3.3.3.1 TASKS AND STEPS

The division of the activities under Task Risk Analysis should be delineated into appropriately sized segments termed 'Task'. A 'Task' may contain general information of the sub activity. Example of 'Tasks' for typical wellhead repair are as follows:

1. Preparation / Rig up.
2. Isolation.
3. Dismantling.
4. Repair / Maintenance activity (main reason for carrying out the work).
5. Reassembly / box up.
6. De-isolation.
7. Clean up / Rig down.

A 'Task' may contain single or multiple steps. 'Step' level may contain sequence of actions to achieve the intent of a 'Task'.

Example of 'Task' and 'Steps' are provided below on Table 4.1.

TABLE 4.1. EXAMPLE OF 'TASK' AND 'STEPS'

Task	Step
Isolation	Trip the well, ensure SSV and SCSSV are closed
	Close the Master Valve
	Close the Casing Valve
	Bleed the annulus pressure until 0 barg, identify any pressure increase (sign of leakage)
	Isolation task completed

3.3.3.2 CONDITIONING FACTORS

Any conditions affecting equipment and piping (i.e. General condition of equipment / piping, Deferrals, Documented Failures / Problems, Temporary repairs / wraps, NRO's (non-routine operations), LTI's (long term inhibits), Outstanding actions from related PHA's, PSSR's and Punch lists) which may impair the robustness of an Intrinsic Control should be identified as a Conditioning Factor.

Given the brownfield nature of Idd El Shargi facilities, the inclusion of Conditioning Factors in the analysis is essential in order to establish the current effectiveness of the controls impacted. A qualified judgment must be made on any Conditioning Factors that affect controls effectiveness on a system and if the outcome is critical, an action item should be recorded, e.g. Long Term Isolation (LTI) on an ESDV will mean it is no longer considered effective in preventing or mitigating an Event Scenario.

Conditioning Factors type is provided in Appendix C of this Guideline Document.

3.3.3.3 HAZARDS IDENTIFICATION

All credible hazards and/ or potential incidents shall be identified. A structured approach will be applied to ensure that all relevant hazards are revealed. The basis for this approach lies in dividing the review scope into a number of Tasks and Steps, and then to do systematic review of each Task and Step to identify the relevant hazards that these systems could be subjected to.

The Project and/ or Change proponent or Task lead will give a brief description of the task Procedure under review in order that the team members to obtain a common understanding of the intent.

During hazards identification for each Task Step, The TRA team shall identify the following:

1. Hazards arising from The Job environment (e.g. SIMOPS, noise, heat exhaustion, etc.).
2. Hazards from the Process Equipment (e.g. high vibration, electrical hazards, radiation, etc.).
3. Hazards related to each Task Step, the team need to brainstorm relevant hazards from 3 main sources while task step is being undertaken:

- Equipment failure,
- External factors,
- Human failure.

Detail of the hazard identification is provided in Appendix F, Idd El Shargi TRA Methodology.

3.3.3.4 CONSEQUENCE ESTIMATION

Since the nature of Task Risk Assessment is non-routine transient activities, the risk matrix used for the TRA is the same risk matrix used in Level 2 Risk Assessment (i.e. ORA, Offshore Risk Assessment) in ISSoW (Integrated Safety System of Work), work process. The detail of this risk matrix is provided in Appendix D, Idd El Shargi Task Risk Assessment Matrix. However, there are differences in how the likelihood is assigned using the Idd El Shargi TRA Risk Matrix as compared to the Level 2 Risk Assessment Matrix.

Consequence shall be analysed before Probability, using the unmitigated, 'worst-case' credible consequence (i.e. reasonable consequences, NOT absolute worst case), for the consequence category in question (e.g. Personnel, Environment, Property Damage and Loss of Revenue, and External impacts).

Consequence should be addressed in the context of Hazard Scenario and Consequence category (i.e. Personnel at Facility, Property Damage/ Loss of Production, Environmental impact), as a series of events progression, and to the point of 'Impact Scenario'. Refer to IS-HSE-GDL-310 for further details of Process Event Scenarios development.

For a consequence category of PD/LOR, consequence should be quantified in terms of financial losses over a period of time. Refer to Idd El Shargi IS-HSE-GDL-316 Appendix B 'What-if' Analysis Guideline, section 4.3.3.4 for Major Hydrocarbon Loss of Containment Scenario consequence rule sets.

3.3.3.5 INTRINSIC CONTROLS

Intrinsic Controls are controls that can be assumed are/ will be in place. In Idd El Shargi TRA method there are several types of intrinsic controls that may be considered robust, they are:

1. Physical controls.
2. Process Controls.
3. Procedural Controls.
4. Human Controls.
5. Timing Controls.
6. Contingency controls.
7. Mitigation controls.

When defining intrinsic controls, the TRA team should aim for the elimination of hazards and reduction of OH&S risks as per the following hierarchy of controls:

a) eliminate the hazard;

b) substitute with less hazardous processes, operations, materials or equipment;

c) use engineering controls and reorganization of work;

d) use administrative controls, including training;

e) use adequate personal protective equipment.

A list of typical controls for a TRA is provided in Appendix E of this Document.

Due to strong reliance on human actions and variability of situations, the controls credited for probability reduction must be:

1. Clearly applicable to the scenario.
2. Reliable based on supporting Key Mitigation Elements (KMEs), and
3. Meet the effectiveness test (i.e. timely and appropriately designed).

3.3.3.6 INITIAL RISK

After consequence level established, 'Intrinsic Controls' need to be identified by TRA team to determine the probability level and risk rank the scenario in question using Idd El Shargi TRA Risk Matrix for the scenario in question. If Initial Risk is not at tolerable level, Additional Controls should be recommended by the team to reduce the risk to ALARP.

The Risk Response to be taken for Risk Levels achieved by the assessment are described below:

1. 15 to 50: Task shall not proceed until it is redefined, substituted, or additional control measures are put into place. Field Operations Team Leader and Operations Manager approval (or above) is required before proceeding following further assessment by a specialist team including the Risk Engineer.
2. 9 to 12: Task may only proceed following Site Managers authorization. Consultations with specialist personnel may be required. Review task to see if risk can be reduced further by application of additional control measures.
3. 1 to 8: No further action required, job may proceed. Risk is tolerable with additional control measures put in place.

3.3.3.7 ADDITIONAL CONTROLS / RECOMMENDATIONS

Change-specific and/or additional robust controls need to be identified and recommended by the TRA team in order to reduce risk from the initial risk levels to ALARP.

Additional Controls shall be categorized accordingly (e.g. Timing, Process, Physical, Contingency, Mitigation, Procedural, and Human Factor), and these categories shall be carefully considered. A list of typical controls for a TRA is provided in Appendix E of this Document.

Emphasis shall be given to robust additional controls with the principles of Inherent safety (e.g. Elimination of the Hazard and the approach of Minimise / Substitute / Moderate / Simplify), where feasible, and Prevention, e.g. reduction of likelihood of the event/ impact scenario.

In addition to specific 'Additional controls' the TRA team should include 'Recommendations'. Note that no credit shall be taken for 'Recommendations' when determining Residual Risk; only 'Additional Controls' with a clear indication that those will be implemented shall be considered in Residual Risk determination.

It is important when determining required actions (Additional controls and recommendations), that indecisive and vague wording is avoided. The item should be recorded such that it can be acted upon by the person designated to execute it. The actions proposed should be feasible, practicable and realistic in the context of risk reduction. The actions may be challenged by the change owner or other responsible parties and this may result in the need for some form of cost benefit analysis.

3.3.3.8 RESIDUAL RISK

The 'Residual Risk' should be determined by the team by means of the modified probability or consequence levels based on the identified Additional Controls, which determine the potentially changed Risk position in the Idd El Shargi TRA Risk Matrix. In the event that no Additional Control for a hazard scenario is identified, the Residual Risk shall be recorded on the worksheet at the same value as the Initial Risk.

Guideline of PHA-Pro TRA worksheet completion is presented in Appendix G.

3.3.3.9 TRA ANALYSIS FLOWCHART

The flowchart in Figure 4.1 below shows the assessment process to be followed.

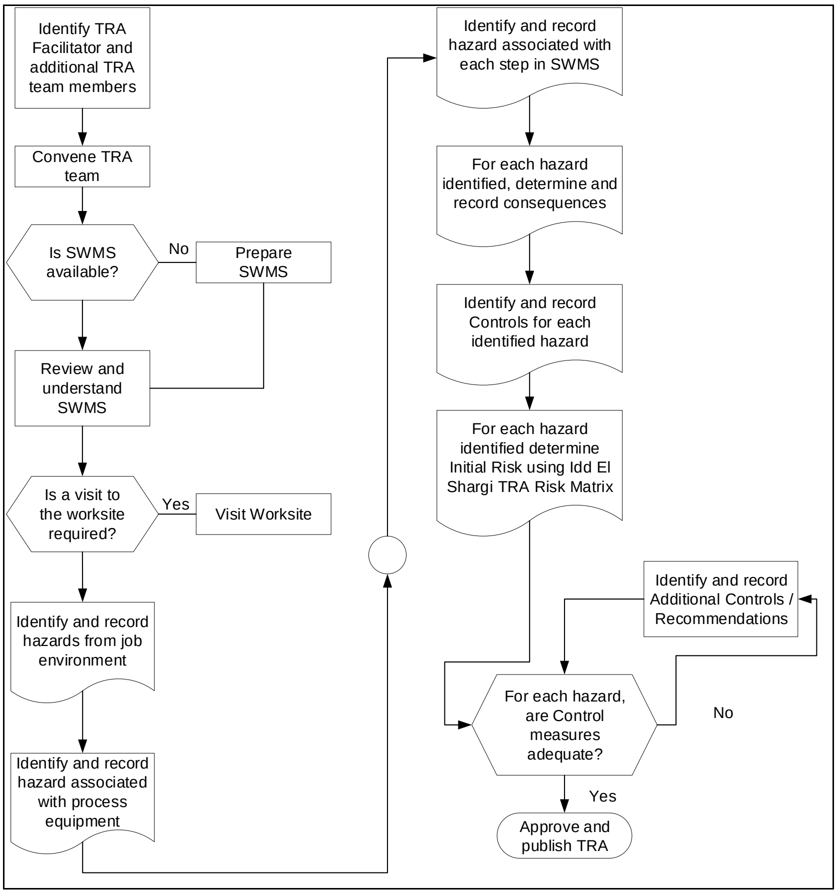


FIGURE 4.1 TRA ANALYSIS FLOWCHART

3.3.4 REPORTING

The TRA analysis report should utilise Standard Idd El Shargi TRA analysis template output report from PHA Pro 8. RM administrator shall issue 'revision A' (e.g. rev. A1, A2, etc. as required), report to all team members for their review and comments. The report content shall be agreed by the team members. Once comments from the team members have been incorporated, RM administrator should issue 'revision B' (e.g. rev.B1, B2, etc. as required), report for approvals and actions completion by action owners. A copy of all reports shall be maintained by RM administrator for record and distribution purposes.

The reports should be issued in line with the requirements of IS-HSE-PRC-301 (Risk Evaluation and Response) and IS-HSE-GDL-303 (RM Administration).

The Standard TRA analysis report should contain:

- A worksheet report (automatically generated by the PHA-Pro software).
- Supporting documentations (e.g. Work method statements, mark-up of P&IDs, plot plans, work pack Documents, etc.).

A completed review report can be used to demonstrate to interested parties that a TRA analysis has been accomplished and all possible actions have been examined and or implemented to eliminate major hazards.

All the drawings, reference data, evaluations, record sheets, action sheets etc., used or developed during the course of the TRA analysis should be filed for future reference.

A typical report will contain a brief description of project or change proposals including Purpose, Scope, and Objective statements, list of team members, dates of meetings, list of documentation being used as basis of review, actions summary and recommendations. Other principal Documents may be presented as an Appendix. The final report is circulated and agreed to by all team members.

4.0 REFERENCES TO OTHER DOCUMENTS

TABLE 1: REFERENCES TO OTHER DOCUMENTS

Document Title	Document Number	Internal/External Document (if any)	Document Referencing
Risk Evaluation and Response Procedure	IS-HSE-PRC-301	Internal	Sideways
Process Hazard Review Procedure	IS-HSE-PRC-308	Internal	Sideways
RM Administration Guideline	IS-HSE-GDL-303	Internal	Sideways
What If Analysis Guideline	IS-HSE-GDL-316	Internal	Sideways
Petroleum and natural gas industries – Offshore production installations Guidelines on tools and	ISO 17776:2000	External	N/A

techniques for hazard identification and risk assessment.			
Occupational health and safety management systems	ISO 45001: 2018	External	N/A

5.0 DEFINITIONS

Refer to the RM Abbreviations & Definitions Guideline, (IS-HSE-GDL-304).

APPENDIX

APPENDIX A - REVISION HISTORY LOG

TABLE 3: REVISION HISTORY LOG

Revision Number: 01 **Date: 31/08/2023**

Item Revised:	Revision Description	Page No.
All	Document simplified as part of QMR initiative.	All
3.3.3.5	Section 3.3.3.5 – added reference to Hierarchy of Controls as per ISO 45001.	9
Remarks:		

APPENDIX B - LIST OF ABBREVIATIONS

Refer to the RM Abbreviations & Definitions Guideline, (IS-HSE-GDL-304).

APPENDIX C - CONDITIONING FACTORS TYPE

Type	Description
General Condition	General Condition of the node
No Known Issues	No Issues
Deferral: • AI-MI (Mechanical Integrity) • AI-Mtce. (Equipment Maintenance)	Inspection etc. (Fabric / Static equipment) Equipment Maintenance; Proof-test etc.
NRO (Non-Routine Operation)	Non-Routine Operation in the node or affecting the node function
Organisational Change	Organisational Change
Procedural Change	Procedural Change
RIK (Replacement-in-Kind)	Replacement in Kind

Type	Description
Temporary Repair	Wraps; Clamps; other- Technical Reviews
LTI (Long-term Isolation)	Long Term Inhibits- ESD, F&G, RV; Firewater
Related MOC's / MOC Requests	Related MOC's - planned or in-hand that may impact the review
Process / Maintenance Overrides	Planned short-term overrides
Historical Incidents	What? When? How? Why? Lessons Learned / Corrective actions?
SIL assessed	Have all the SIF (Safety Instrumented Functions) loops been SIL assessed and verified in the node?
No FTI	Is all key Facilities Technical Information available and as-built for the node?
PHA-Actions Outstanding	PHA Actions which have not yet been implemented: FEC; OPS; Especially JSA-transient works
Punch List Outstanding	Construction e.g. Temporary (wrong-specification material) bolts fitted; Cat. A; Cat. B
Procedural Issue	Procedure (NI; NC). Operations - chronic non-conformance or Procedure that does not exist or needs improvement
CAR (Corrective Action Request)	Audit; Incident - Various depts. All these - once accepted, if not actioned, constitute a subtle degradation of MOC
No FTI	e.g. Weather; Stress; Competencies
Deficiency: • AI-MI (Mechanical Integrity) • AI-Mtce. (Equipment Mtce.)	< MAWT; Leak; Structural Damage Equipment / System Performance; Poor condition; Blocked RV / Impulse Lines; Difficult to maintain; 'Ex' eqpmt. condition
Operating Phase	Change from Normal Ops., e.g. Start-up, shutdown, Emergency shutdown; Black start

Type	Description
Coincident Works	Simops; Triops
Defect Design	BV or Instrument access; Escape Routes compromised; Dead-legs; Risers External to jacket
Personnel in Area	Personnel in Area

APPENDIX D - IDD EL SHARGI TRA RISK MATRIX

Likelihood (Over Duration of Task	IDD EL SHARGI TASK RISK ASSESSMENT MATRIX						
Very High (H) - Almost definite that a hazard will be realized (no Controls in place)	Likelihood	10	10	20	30	40	50
High (H) – Very likely that a hazard will be realized if a barrier fails (at least 1 Control in place)		5	5	10	15	20	25
Medium (M) – Moderately likely that a hazard will be realized if barriers fail (at least 2 independent Controls in place)		3	3	6	9	12	15
Low (L) – Unlikely that a hazard will be realized if barriers fail (at least 3 independent Controls in place)		2	2	4	6	8	10
Very Low (VL) – Very unlikely that a hazard will be realized if barriers fail (at least 4 independent Controls in place)		1	1	2	3	4	5
			1	2	3	4	5
			Consequence				

	1	2	3	4	5
Personnel injury	First Aid Injury	Lost time injury, medical treatment, Onshore treatment by doctor	Hospital stay and substantial lost time	One (1) permanently incapacitating injuries	More than 1 permanent incapacitating injury
Property Damage / Production Impact	Minimal Loss <$75K or <1,500 barrels	Minor Loss $75K to $250K or 1,500 to 5,000 barrels	Moderate Loss $250K to $500K or 5,000 to 10,000 barrels	Significant Loss $500K to $1MM, or 10,000 to 20,000 barrels	Major Loss >$1MM or >20,000 barrels
Environmental Impact	Slight loss of containment <1 barrel	Minor loss of containment < 10 barrels, only the work place affected	Significant loss of containment >100 barrels, release controllable from station	Significant loss of containment >100 barrels but controllable within Idd El Shargi	Total loss of containment and requiring outside agencies

Risk Level	Risk Response
15 to 50	Task shall not proceed until it is redefined, substituted, or additional control measures are put into place. Field Operations Team Leader and Operations Manager approval (or above) is required before proceeding following further assessment by a specialist team including the Risk Engineer.
9 to 12	Task may only proceed following Site Managers authorization. Consultations with specialist personnel may be required. Review task to see if risk can be reduced further by application of additional control measures.
1 to 8	No further action required, job may proceed. Risk is tolerable with additional control measures put in place.

NOTE: Generally, non-routine transient activities should be covered by the Idd El Shargi TRA Matrix. For routine transient jobs/activities (e.g. drilling, well servicing, transportation etc.), the standard Idd El Shargi Risk Matrix must be used to comply with the HSE Risk Evaluation & Response Procedure (QP-ISND-RM-HSE-308).

APPENDIX E - TYPICAL CONTROL FOR TASKS

Physical
- ❑ Properly rated spade or blank flange
- ❑ Removal of pipe-work or fuses
- ❑ Equipment isolation via LOTO
- ❑ Mechanical barrier
- ❑ Locked enclosure
- ❑ Keep personnel at distance
- ❑ Eliminate or substitute toxic substances
- ❑ Substitute noisy machinery
- ❑ Use mechanical handling equipment to avoid injury
- ❑ Use of certified equipment (e.g. scaffold, lifting gear, test equipment)

Process
- ❑ Process controls
- ❑ Alarms
- ❑ Shutdowns
- ❑ Isolation valves
- ❑ Pressure relief systems

Procedural
- ❑ Test for pressure build-up or leaks
- ❑ Examination of flushing fluids
- ❑ Test for toxic atmosphere or flammable gas
- ❑ Procedure for control of SIMOPS and adjacent work
- ❑ Prohibition of hot work
- ❑ Critical lift plan
- ❑ Established contingency plan (applicable to PD-LOR only)
- ❑ Heat stress policy
- ❑ Formalized standing instructions

Human
- ❑ Use of consultants / contractors or specialist personnel
- ❑ Continuous supervision
- ❑ Verification of steps by independent personnel

Timing
- ❑ Limit duration of task
- ❑ Use lower risk window of time
- ❑ Documented/check listed worksite preparation and planning for movement of materials, tools, etc.

Contingency
- ❑ Reduction of inventory
- ❑ Spare equipment
- ❑ Deluge and blown-down systems
- ❑ Remote shutdown capability

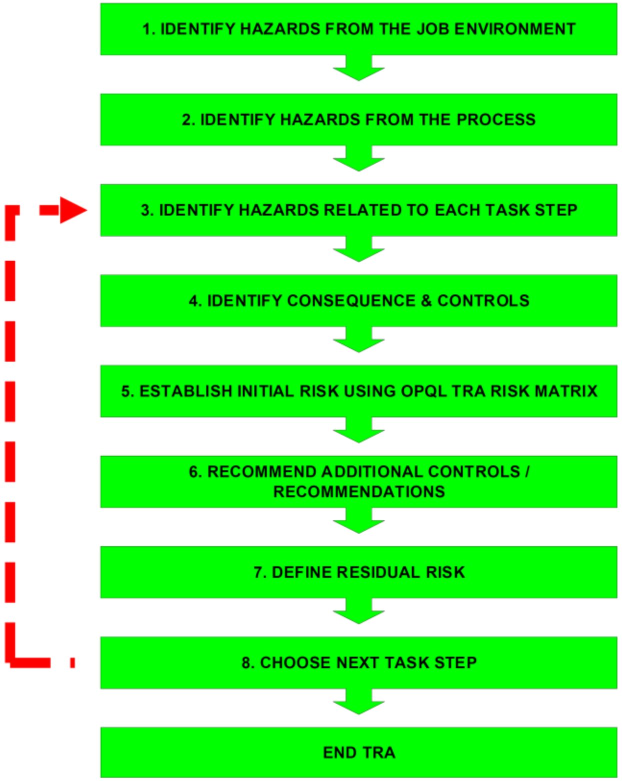
Mitigation
- ❑ Temporary refuge
- ❑ Fire/blast wall
- ❑ Water curtain
- ❑ Provision of PPE and clear escape paths
- ❑ Specialized PPE
- ❑ Fall protection (e.g. safety harness)

IMPORTANT

Due to strong reliance on human action and variability of situations, the Controls credited must be:
1. Clearly applicable to the scenario;
2. Reliable based on supporting **key mitigation elements (KMEs)**; and
3. Meet the effectiveness test (i.e. timely & appropriately designed).

NOTES
- Other controls may be identified by the TRA Team and must be confirmed with the
- Only Controls are given for risk reduction based on application to scenario and supporting KMEs Idd El Shargi Risk Engineer
- KMEs are activities, specifications and design features that support and maintain the reliability of the Intrinsic / Additional Controls
- In the context of TRAs, KMEs typically include:
  - Routine training
  - Normal testing & inspection
  - Maintenance
  - Communication
  - Signage

APPENDIX F - IDD EL SHARGI TYPICAL METHODOLOGY



The TRA Facilitator is responsible for assembling relevant data including:

- Safe Work Method Statement
- Plot plans, General Arrangement & Elevation Drawings
- Site Survey document (e.g. photgraphs)
- Material Data Sheets
- Isolation Pack
- Procedures and work instructions
- PFD, P&IDs, etc.

Step 1 – Identify Hazards from the Job Environment
Credible hazards from the following main categories can be explored:

- ❑ Simultaneous Operations (SIMOPS)
- ❑ Noise
- ❑ Heat exhaustion
- ❑ Slips and trips
- ❑ Working at heights
- ❑ Confined spaces
- ❑ Dropped or falling hazards
- ❑ Distractions
- ❑ Nearby hot works / grinding / bolting
- ❑ Nuisance odours
- ❑ Overhead power lines
- ❑ Ventilation / air quality
- ❑ Poor visibility / lighting
- ❑ Posture / clearance / strain
- ❑ Moving vehicle hazards
- ❑ Hand hazards
- ❑ Asphyxiation
- ❑ Unguarded openings
- ❑ Restricted egress
- ❑ etc.

Step 2 – Identify Hazards from Process Equipment
Credible hazards from the following main categories can be explored:

- ❑ Chemical spills
- ❑ High vibration
- ❑ Electrical hazards
- ❑ Moving machinery
- ❑ Vents / drains
- ❑ Relief system discharges
- ❑ High pressure hazards
- ❑ Wax / sand / sludge / scale
- ❑ Hazardous waste
- ❑ Radiation
- ❑ Heat sources
- ❑ Asbestos
- ❑ Ignition sources
- ❑ Static discharge potential
- ❑ etc.

Step 3 – Identify Hazards Related to Each Task Step
Team to brainstorm hazards from 3 main sources while task step is being undertaken:

1) Equipment failure,
2) External factors
3) Human failure

The main focus will be on human failure as the human element is a large part of a non-routine task completion. Credible hazards from the following main categories can be explored:

- ❑ Omission errors (action is not performed)
- ❑ Commission errors (action is performed incorrectly)
- ❑ Extraneous act (non-required action is performed instead of or in addition to required act)
- ❑ Violation or deliberate acts (Action that is prohibited, or different from the prescribed)

Other important human failure guide words:

- ❑ Early (action is performed earlier than specified)
- ❑ Late (action is performed later than specified)
- ❑ Out of sequence (action is performed in the wrong order)

APPENDIX G - SAMPLE TRA ANALYSIS WORKSHEETS

Worksheet Completion Guidance